IBM BOB HACKATHON 2026 / EXPERIENCED DEVELOPERS: MODERNIZE WHAT MATTERS

Bob Hackathon Control Plane

One private control room for a developer's work, service health and test proof. It runs only on your machine, and IBM Bob modernized every layer of it.

77/77 tests passing | **0** package installs | **127.0.0.1** only | **6** Bob agent tasks

KARIIM CHILES, SOLO BUILDER

TEAM AND PROJECT

The view small teams don't have

The problem

- Tasks, health checks and test proof live in five different tools.
- The cloud dashboards that join them up copy private machine data off the laptop.
- Nobody can show, in one place, that the work is done and proven.

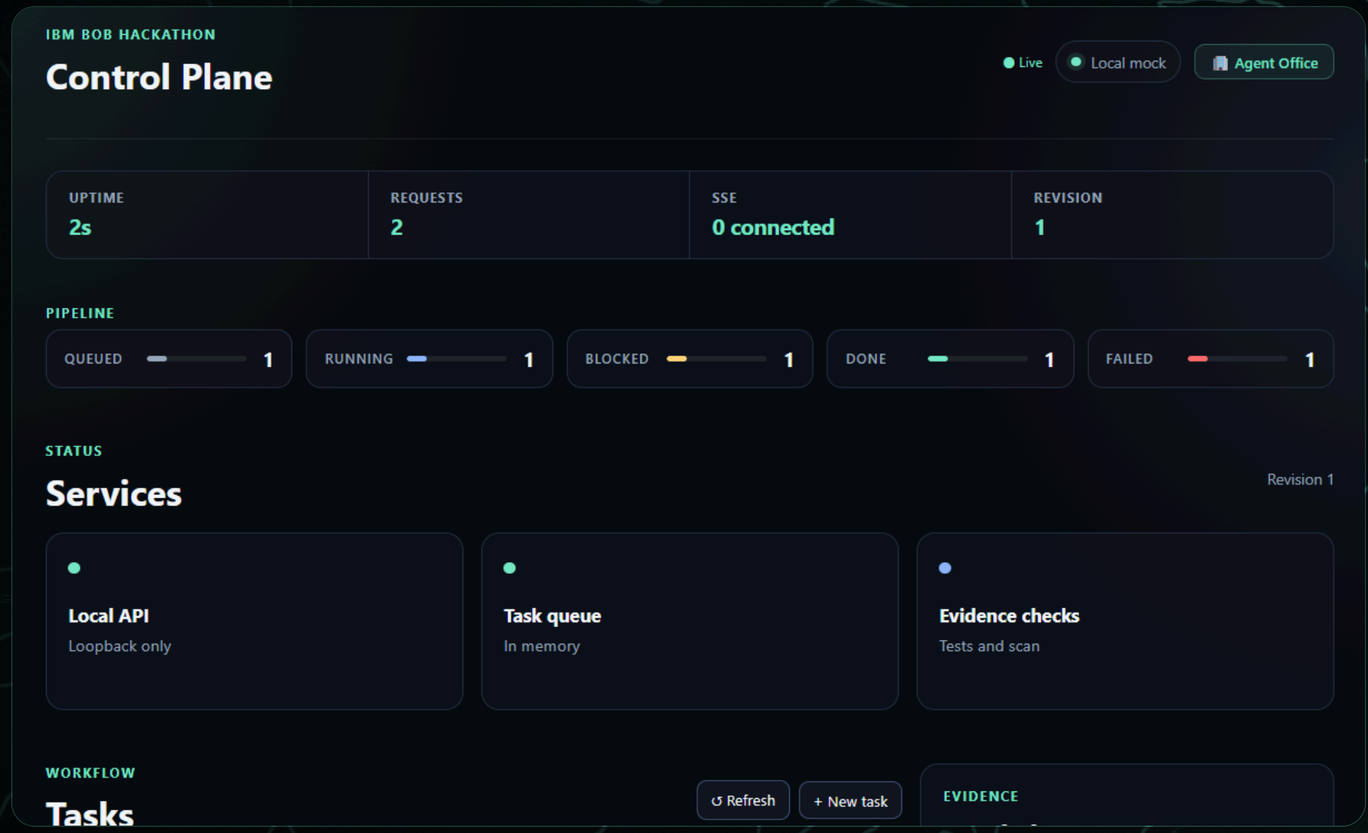
The project

TEAM	Kariim Chiles, solo
TRACK	Experienced Developers: Modernize What Matters
BUILT WITH	IBM Bob IDE, Node.js 22 standard library, plain HTML, CSS and JavaScript, Three.js
BOUNDARY	Loopback only, mock data, no outside calls. The first commit is the pre-event baseline; only Bob's event work is claimed.

THE SOLUTION / 1 OF 2

A live control room

- Pipeline rail counts every task as queued, running, blocked, done or failed.
- Live push sends task updates through a server event stream.
- Task workflow: create, advance, block or fail with a reason, reset, delete.
- Activity log keeps the evidence trail; metrics show uptime and load.



THE SOLUTION / 2 OF 2

The Agent Office, in 3D

- Every task is a character at a desk in a live 3D room.
- Its status rides above its head and changes the moment the work changes.
- Same event stream as the dashboard, so the room never lies.
- Rendered on the GPU with Three.js, served locally, works offline.



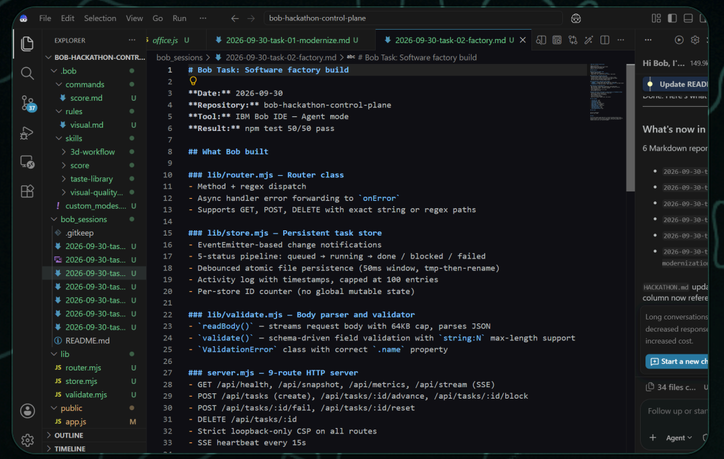
HOW IBM BOB BUILT IT

Six agent-mode tasks, each one proven

T1	Modernize the baseline Server error logging, graceful shutdown, stuck-button fix	13 TESTS
T2	Software factory build Router, persistent store, validator, event stream, dashboard overhaul	50 TESTS
T3	Visual quality system Quality-gate skill, taste library skill, visual rules, /score command	50 TESTS
T4	3D workflow Eight-phase 3D production skill, mesh validator, shader library	50 TESTS
T5	Bug fixes Promise leak in the router, unsafe HTML pattern, reconnect backoff	52 TESTS
T6	Full system modernization Agent Office, store and server hardening, strict security policy	77 TESTS

Bob was also taught our standards, kept in the repo for every later task:

```
.bob/custom_modes.yaml
.bob/skills/ (4 skills)
.bob/rules/visual.md
.bob/commands/score.md
```



FUNCTIONALITY AND TECHNICAL EXECUTION

Proof, not promises

```
$ npm test
✓ POST /api/tasks creates a new task
✓ POST /api/tasks/:id/block marks task as blocked
✓ POST /api/tasks/:id/advance returns 409 for wrong-state task
✓ DELETE /api/tasks/:id returns 404 for unknown task
ℹ tests 77   ℹ pass 77   ℹ fail 0

$ npm run scan
PRIVACY SCAN PASS
```

77
tests, all passing

12
API endpoints

0
package installs

1
address: 127.0.0.1

RUN ON THE FINAL WORKING COPY, 17 OCTOBER 2026

AGAINST THE JUDGING CRITERIA

Where the points come from

30%

IBM Bob usage

- Six agent-mode tasks across server, store, client, 3D and fixes.
- Custom modes, four skills, rules and a command written into `.bob/`.
- Report and screenshot saved for every task.

30%

Innovation

- Private by design: one view of work, health and proof that never leaves the machine.
- Work you can see: tasks become people in a live 3D room.
- Bob turned into a reviewer with its own quality gate.

30%

Execution

- 77 of 77 tests and a privacy scan pass.
- 12 endpoints, live push, file persistence, input checks, 404 / 409 / 422 errors.
- Server uses the Node standard library. No install step.

10%

Presentation

- Two commands to run it.
- Demo under three minutes.
- Every claim in this deck has a test, a file or a screenshot behind it.

KEY LEARNINGS

What working with Bob taught us

L1 One big, clear task beats ten small ones

The software factory task touched eleven files and took the suite from 13 to 50 tests in one pass, because the goal and the finish line were stated up front.

L2 Write the standards down once

Rules, skills and a scoring command in `.bob/` meant every later task started with the same bar instead of a fresh explanation.

L3 Tests are the receipt

No task counted until the suite passed. The count is the progress log: 13, 50, 52, 77.

L4 Privacy needs a machine check

A promise not to leak data is worth nothing. A scan that runs before every commit is worth a lot.

IMPROVEMENT OPPORTUNITIES AND ROADMAP

What comes next

Open improvements

- Move the last inline styles into stylesheets so the strict security policy never blocks them.
- Keyboard and screen-reader pass on the 3D office.
- Evidence for more than one team member.

NEXT

Real local signals

Git status, test runs and service checks from the same machine, behind the same loopback wall.

THEN

Bob drives the pipeline

Bob tasks create and advance control-plane tasks directly, with their reports attached.

LATER

Team mode

Several machines, one shared view, and a signed evidence export for reviews and audits.

TRY IT

Local. Private. Proven.

```
1 npm start
```

```
2 open http://127.0.0.1:8787
```

Node.js 22 or newer. Nothing to install. `npm test` and `npm run scan` prove it.